

Lecture 7

September 15

Learning objectives

- Derive $\text{Var}(Y_1 + Y_2)$, and explain why variances do not generally add.
- Compare the variance of $(Y_1 + Y_2)$ when the variables are independent and when $Y_2 = Y_1$.
- Calculate the variance of binomial random variable
- Explain how Simpson used an assumed distribution of measurement errors to assess the accuracy of an average.
- Explain Bayes's criticism of Simpson's assumptions using an imperfect instrument; distinguish reduced random variation from systematic bias, and explain why averaging does not necessarily bring measurements closer to the truth.

Last time:

Expectation: $E[Y] = \sum_y y P(Y=y)$

$$E[Y_1 + Y_2 + \dots + Y_n] = E[Y_1] + \dots + E[Y_n]$$

Variance:

$$\text{Var}(Y) = E[(Y - E[Y])^2] = E[Y^2] - (E[Y])^2$$

Examples:

For **Bernoulli** R.V. Y : $E[Y] = p$, $\text{Var}(Y) = p(1-p)$

$Y_1, \dots, Y_n$ - indep. Bernoulli(p), then $X = Y_1 + \dots + Y_n \sim \text{Binom}(n, p)$
 $E[X] = np$

Variance of $X \sim \text{Binom}(n, p)$:

$$\text{Var}(X) = E[(X - E(X))^2]$$

$$= E[X^2] - (E[X])^2$$

$$E[X] = np \quad (E[X])^2 = n^2 p^2$$

$$E[X^2] = \sum_{k=0}^n k^2 P(X=k)$$

~~$$E[X] = \sum_{k=0}^n k P(X=k) = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}$$~~

$$E[X] = E[Y_1 + \dots + Y_n] = E[Y_1] + \dots + E[Y_n] = np$$

$$E[X^2] = E[(Y_1 + \dots + Y_n)^2]$$

For $n = 2$:

$$X = Y_1 + Y_2, \quad Y_1, Y_2 \text{ indep. Bern}(p)$$

$Y_2 \backslash Y_1$	0	1
0	$(1-p)^2$	$p(1-p)$
1	$(1-p)p$	p^2

$$p(Y_1 = 0, Y_2 = 0)$$

$$X = \begin{cases} 0 & \text{w.p. } (1-p)^2 \\ 1 & \text{w.p. } 2p(1-p) \\ 2 & \text{w.p. } p^2 \end{cases}$$

$$X = \begin{cases} 0 & \text{w.p. } (1-p)^2 \\ 1 & \text{w.p. } 2p(1-p) \\ 2 & \text{w.p. } p^2 \end{cases}$$

$$E[X^2] = \sum_{k=0}^2 k^2 P(X=k)$$

$$= 0^2 \cdot (1-p)^2 + 1^2 \cdot 2p(1-p) + 2^2 p^2$$

$$= 2p - 2p^2 + 4p^2 = 2p + 2p^2$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 2p + 2p^2 - 4p^2$$

$$= 2p - 2p^2 = 2p(1-p)$$

Is variance linear?

$n=2$

$$\text{Var}(X) = \text{Var}(Y_1 + Y_2) = 2p(1-p)$$

$$\text{Var}(Y_1) + \text{Var}(Y_2) = p(1-p) + p(1-p) = 2p(1-p)$$

In this case - yes?

What if $Y_1 = Y_2$

$$\text{Var}(Y_1 + Y_2) = \text{Var}(Y_1 + Y_1) = \text{Var}(2Y_1) = 4p(1-p)$$

On Hw, you'll show $\text{Var}(aY) = a^2 \text{Var}(Y)$

not independent

If $Y_1, \dots, Y_n$ are independent, then

$$\text{Var}(Y_1 + \dots + Y_n) = \text{Var}(Y_1) + \text{Var}(Y_2) + \dots + \text{Var}(Y_n)$$

Binomial: $X = Y_1 + \dots + Y_n$, $Y_k \overset{\text{indep}}{\sim} \text{Bern}(p)$

$$\text{Var}(Y_k) = p(1-p)$$

$$\text{Var}(X) = \text{Var}(Y_1 + \dots + Y_n) = \underbrace{\text{Var}(Y_1) + \dots + \text{Var}(Y_n)}_{p(1-p)}$$

$$\text{Var}(X) = n \text{Var}(Y_1) = np(1-p)$$

$$\text{SD}(X) = \sqrt{np(1-p)}$$

If Y_1 is not indep. of Y_2 ?

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$\text{Var}(Y_1 + Y_2) = \text{Var}(Y_1) + \text{Var}(Y_2) + 2\text{cov}(Y_1, Y_2)$$



"covariance"

$$\text{Var}\left(\sum_{k=1}^n Y_k\right) = \sum_{k=1}^n \text{Var}(Y_k) + \sum_{k \neq j} \text{cov}(Y_k, Y_j)$$

De Moivre : $\hat{p} \in (p - \#, p + \#)$

or $P(\text{data} | \text{parameter})$

For $X \sim \text{Binomial}(n, \frac{1}{2})$

$$P(X = \frac{n}{2} + \ell) \approx \frac{2}{\sqrt{n \cdot 2\pi}} \exp\left(-\frac{\ell^2}{n/2}\right)$$

We want : $p \in (\hat{p} - \#, \hat{p} + \#)$ or $P(\text{parameter} | \text{data})$

Say something about p from $\hat{p}$

Simpson: justify averaging observations

Measurement: $M_i = \underset{\text{truth}}{\theta} + \epsilon_i \rightarrow M_i - \theta = \epsilon_i$

The error of the average = average of the errors

$$\bar{M} - \theta = \bar{\epsilon}$$

Reasonable assumptions:

- symmetric around truth
- small errors are more common
- limits of errors

Suppose each measurement has independent error with the following distribution:

$$\epsilon_i = \begin{cases} -1 & \text{w.p. } 1/4 \\ 0 & \text{w.p. } 1/2 \\ 1 & \text{w.p. } 1/4 \end{cases}$$

$$\begin{matrix} r^{-v}, & 2r^0, & r^v \\ \downarrow & \downarrow & \downarrow \\ 1, & 2, & 1 \end{matrix} \quad \begin{matrix} r=1 \\ v=1 \\ \end{matrix}$$

$$[1, 2, 1] = 4$$

What assumptions does this model make?

- Symmetry
- 0 is more likely

$$\varepsilon = \begin{cases} -1 & \text{w.p. } \frac{1}{4} \\ 0 & \text{w.p. } \frac{1}{2} \\ 1 & \text{w.p. } \frac{1}{4} \end{cases}$$

$$\underline{P(|\varepsilon| \leq \frac{1}{2}) = P(\varepsilon = 0) = \frac{1}{2}}$$

$$P(|\varepsilon_1 + \varepsilon_2| \leq \frac{1}{2}) = P(|\varepsilon_1 + \varepsilon_2| = 0)$$

How can we have $|\varepsilon_1 + \varepsilon_2| \leq \frac{1}{2}$

$\varepsilon_1 + \varepsilon_2$ can be $-2, -1, 0, 1, 2$

When is $\varepsilon_1 + \varepsilon_2 = 0$: $(0, 0), (-1, 1), (1, -1)$

$$P(\varepsilon_1 + \varepsilon_2 = 0) = P(\varepsilon_1 = 0, \varepsilon_2 = 0) + P(\varepsilon_1 = -1, \varepsilon_2 = 1) + P(\varepsilon_1 = 1, \varepsilon_2 = -1)$$

$$= \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{6}{16} = \frac{3}{8}$$

$$\epsilon = \begin{cases} -1 & \text{w.p. } \frac{1}{4} \\ 0 & \text{w.p. } \frac{1}{2} \\ 1 & \text{w.p. } \frac{1}{4} \end{cases}$$

$\epsilon_1 + \epsilon_2$ can be $-2, -1, 0, 1, 2$

$$P\left(\left|\frac{\epsilon_1 + \epsilon_2}{2}\right| \leq \frac{1}{2}\right) = P(|\epsilon_1 + \epsilon_2| \leq 1)$$

$$= 1 - P(\text{sum} = 2 \text{ or } -2)$$

$$P(|\epsilon_1 + \epsilon_2| = 2) = P(\epsilon_1 = 1, \epsilon_2 = 1) + P(\epsilon_1 = -1, \epsilon_2 = -1)$$

$$= \frac{1}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{2}{16}$$

$$P\left(\left|\frac{\epsilon_1 + \epsilon_2}{2}\right| \leq \frac{1}{2}\right) = 1 - \frac{2}{16} = \frac{14}{16} = \frac{7}{8} > \frac{1}{2}$$

2 obs						1 obs		
sum	-2	-1	0	1	2	-1	0	1
avg error	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	-1	0	1
prob.	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

What if we average more than 2 errors?

Can $P(|\bar{E}| \leq \frac{1}{2}) = 1$ for any fixed n ?

Bayes: averaging - good

but what if there is a systematic error?

Ex: A clock is always within 2 seconds of correct time.

If we average many measurements with it, must the error approach zero?

Simpson revises

1. That there is nothing in the construction, or position of the instrument whereby the errors are constantly made to tend the same way, but that the respective chances for their happening in excess, and in defect, are either accurately, or nearly, the same.
2. That there are certain assignable limits between which all these errors may be supposed to fall; which limits depend on the goodness of the instrument and the skill of the observer.

(Simpson, 1757, p. 64)